



Lab Title: Virtual Lab Environment Setup, Connectivity Verification & Web Proxy Configuration

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Course Name: BVWS102 – OWASP Top 10 Vulnerabilities & Exploitation Techniques

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Executive Summary

This lab assessment documents the successful setup and validation of a basic offensive security virtual laboratory using industry-standard tools and intentionally vulnerable systems. As a cybersecurity student, I configured a controlled lab environment consisting of Kali Linux as the attacker machine and Metasploitable2 as the target system using a virtualization platform. Network connectivity between both machines was verified to ensure proper communication within the lab environment.

Furthermore, a vulnerable web application (Damn Vulnerable Web Application – DVWA) hosted on the Metasploitable server was accessed from the attacker machine to confirm service availability. Burp Suite Professional was installed, licensed, and configured alongside the Firefox browser using FoxyProxy to enable web traffic interception. This setup allowed for real-time analysis of HTTP requests and responses, simulating a real-world web application penetration testing workflow.

Overall, this lab demonstrates my understanding of virtual lab deployment, attacker-target communication, vulnerable application testing, and professional web proxy configuration, which are essential foundational skills in offensive cybersecurity and ethical hacking.

By completing these tasks, the lab reinforces practical skills required for web application security testing, vulnerability analysis, and penetration testing in a controlled and ethical environment.

Lab Objective

The primary objective of this lab was to assess my ability, as a cybersecurity student, to design, configure, and validate a functional offensive security testing environment. Specifically, the lab aimed to ensure that I could:

- Properly install and run Metasploitable2 as a vulnerable target machine
- Verify network connectivity between an attacker (Kali Linux) and a target system
- Deploy and access a vulnerable web application using the target's IP address
- Install and license Burp Suite Professional correctly
- Configure browser proxy settings to intercept and analyze web traffic

By completing these tasks, the lab reinforces practical skills required for web application security testing, vulnerability analysis, and penetration testing in a controlled and ethical environment.

Tools Used

Hypervisor – Vmware Workstation – Hosting and managing virtual machines.

Attacker machine – Kali linux – Performing penetration testing activities.

Target Machine – Metasploitable2 – Internationally vulnerable system for testing.

Web Browser – Mozilla Firefox (kali Linux) – Accessing vulnerable web applications.

Proxy Tool – Burp Suite Professional – Intercepting and analyzing web traffic.

Browser Addon – FoxyProxy – Managing proxy configuration easily

Vulnerable Web App – DVWA (Damn Vulnerable Web Application – Practicing web application exploitation.

Methodology

The lab was carried out using a structured, step-by-step methodology aligned with standard penetration testing practices.

First, Metasploitable2 was imported into the hypervisor and configured using a Host-Only / NAT networking mode to ensure isolation while allowing communication with the Kali Linux attacker machine. The successful boot of Metasploitable confirmed that the target system was operational.

Next, network connectivity was verified by identifying the IP address assigned to Metasploitable2 and sending ICMP echo requests (ping) from Kali Linux. Successful responses confirmed that both machines were on the same network and capable of communication.

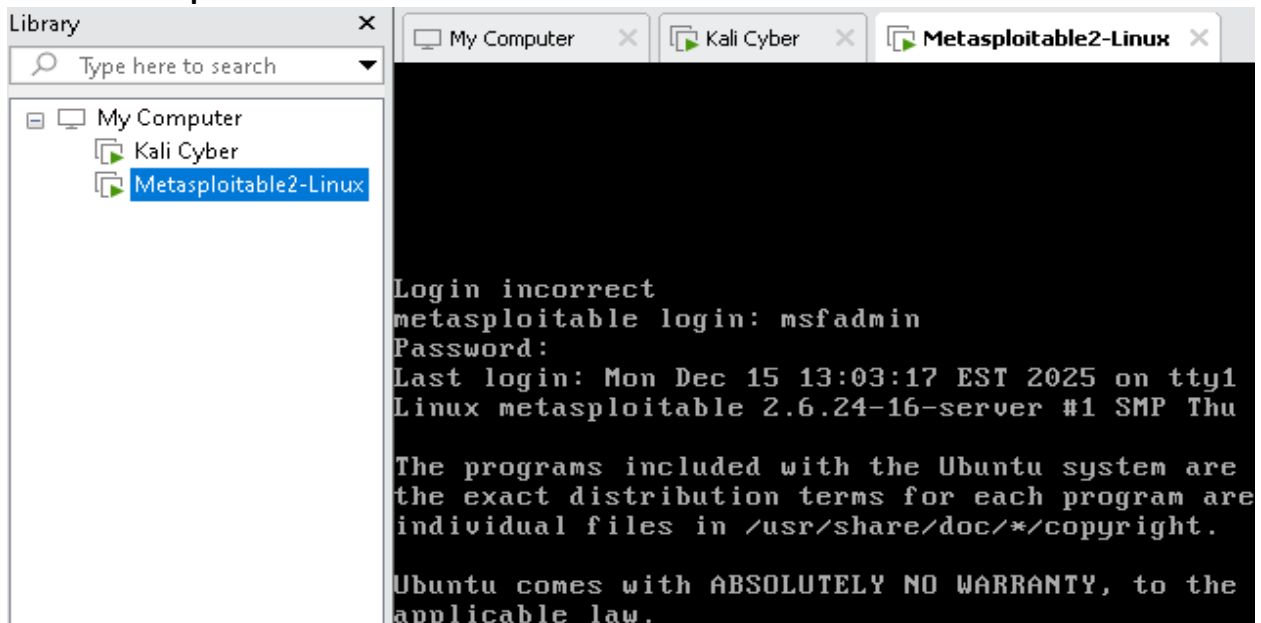
After connectivity verification, the vulnerable web application (DVWA) hosted on Metasploitable was accessed through the Firefox browser on Kali Linux using the target's IP address. This confirmed that the web service was running and reachable.

Burp Suite Professional was then installed and licensed using the approved method demonstrated during class sessions. Proper licensing was confirmed by verifying the absence of trial limitations or expiration warnings.

Finally, Firefox was configured with the FoxyProxy extension to route browser traffic through Burp Suite. Interception mode was enabled within Burp, and HTTP requests generated from the browser were successfully captured and paused. This demonstrated effective proxy configuration and the ability to inspect web traffic, which is a critical skill in web application security testing.

Lab Task

Task 1: Metasploitable Installation



The screenshot shows a Metasploit Meterpreter session with three tabs: 'My Computer', 'Kali Cyber', and 'Metasploitable2-Linux'. The 'Metasploitable2-Linux' tab is active, displaying a terminal window. The terminal output shows a failed login attempt for the 'msfadmin' user. The message 'Login incorrect' is displayed, followed by 'metasploitable login: msfadmin' and 'Password:'. Below this, the system logs the last login as 'Mon Dec 15 13:03:17 EST 2025 on tty1' and shows the system version 'Linux metasploitable 2.6.24-16-server #1 SMP Thu'. The terminal also displays the standard Ubuntu disclaimer about the programs included with the system and the warranty disclaimer.

```
Library
Type here to search

My Computer
  Kali Cyber
  Metasploitable2-Linux

Login incorrect
metasploitable login: msfadmin
Password:
Last login: Mon Dec 15 13:03:17 EST 2025 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu

The programs included with the Ubuntu system are
the exact distribution terms for each program are
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the
applicable law.
```

```
Login incorrect
metasploitable login: msfadmin
Password:
Last login: Mon Dec 15 13:03:17 EST 2025 on tty1
Linux metasploitable 2.6.24-16-server #1 SMP Thu Apr 10 13:58:00 UTC 2008 i686

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ echo hey
hey
msfadmin@metasploitable:~$ print hello world
Warning: unknown mime-type for "hello" -- using "application/*"
Warning: unknown mime-type for "world" -- using "application/*"
Error: no "print" mailcap rules found for type "application/*"
Error: no such file "world"
msfadmin@metasploitable:~$ _
```

Network Address Translation (NAT) is a networking method that allows virtual machines to access external networks, such as the internet, using the host system's IP address while remaining isolated from direct inbound connections. In a virtual lab environment, NAT enables the attacker and target machines to communicate with each other and reach external resources without being directly exposed to the host network. This provides a controlled and secure setup that is suitable for penetration testing and cybersecurity training labs.

My Computer

Kali Cyber

Metasploitable2-Linux

Metasploitable2-Linux

▶ Resume this virtual machine

🔧 Edit virtual machine settings

▼ Devices

Memory	512 MB
Processors	1
Hard Disk (SCSI)	8 GB
CD/DVD (IDE)	Auto detect
Network Adapter	NAT
Network Adapter 2	Host-only
USB Controller	Present
Display	Auto detect

▼ Description

This is Metasploitable2 (Linux)

Metasploitable is an intentionally vulnerable Linux

```
Login incorrect
metasploitable login: msfadmin
Password:
Last login: Mon Dec 15 13:03:10
Linux metasploitable 2.6.24-10

The programs included with the
the exact distribution terms for
individual files in /usr/share

Ubuntu comes with ABSOLUTELY NO
applicable law.

To access official Ubuntu docu
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ cd /
msfadmin@metasploitable:~$ pr
Warning: unknown mime-type for
Warning: unknown mime-type for
Error: no "print" mailcap rule
Error: no such file "world"
msfadmin@metasploitable:~$
```

▼ Virtual Machine Details

State:

Configuration file:

Hardware compatibility:

Primary IP address:

Task 2: Network Connectivity Verification

- Screenshot of Metasploitable IP address

```
Error: no "print" mailcap rules found for type "application/*"
Error: no such file "world"
msfadmin@metasploitable:~$ ipconfig
-bash: ipconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:75:c5:8b
          inet addr:192.168.146.129  Bcast:192.168.146.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe75:c58b/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:924 errors:0 dropped:0 overruns:0 frame:0
          TX packets:123 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:65342 (63.8 KB)  TX bytes:18813 (18.3 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:522 errors:0 dropped:0 overruns:0 frame:0
          TX packets:522 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:223201 (217.9 KB)  TX bytes:223201 (217.9 KB)

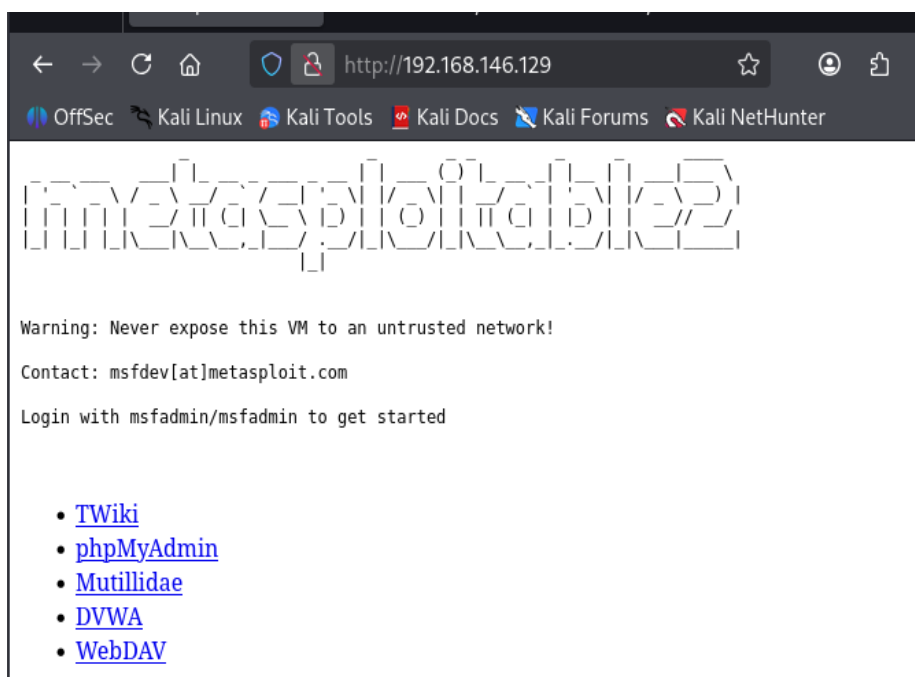
msfadmin@metasploitable:~$ _
```

- Screenshot of successful ping command from Kali Linux to Metasploitable

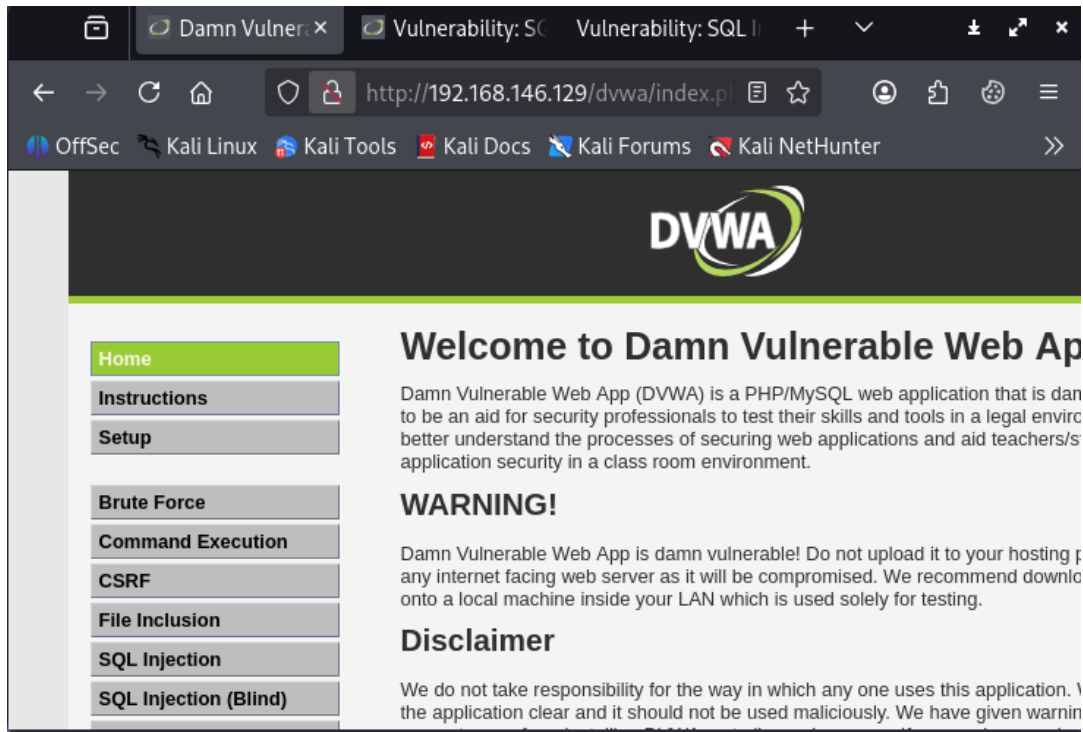
```
(raph@kali)-[~]
$ ping 192.168.146.129
PING 192.168.146.129 (192.168.146.129) 56(84) bytes of data.
64 bytes from 192.168.146.129: icmp_seq=1 ttl=64 time=0.908 ms
64 bytes from 192.168.146.129: icmp_seq=2 ttl=64 time=3.14 ms
64 bytes from 192.168.146.129: icmp_seq=3 ttl=64 time=1.28 ms
64 bytes from 192.168.146.129: icmp_seq=4 ttl=64 time=1.13 ms
64 bytes from 192.168.146.129: icmp_seq=5 ttl=64 time=0.953 ms
64 bytes from 192.168.146.129: icmp_seq=6 ttl=64 time=0.465 ms
64 bytes from 192.168.146.129: icmp_seq=7 ttl=64 time=0.873 ms
64 bytes from 192.168.146.129: icmp_seq=8 ttl=64 time=1.16 ms
64 bytes from 192.168.146.129: icmp_seq=9 ttl=64 time=1.18 ms
64 bytes from 192.168.146.129: icmp_seq=10 ttl=64 time=0.972 ms
64 bytes from 192.168.146.129: icmp_seq=11 ttl=64 time=0.674 ms
64 bytes from 192.168.146.129: icmp_seq=12 ttl=64 time=0.843 ms
64 bytes from 192.168.146.129: icmp_seq=13 ttl=64 time=0.981 ms
64 bytes from 192.168.146.129: icmp_seq=14 ttl=64 time=1.38 ms
64 bytes from 192.168.146.129: icmp_seq=15 ttl=64 time=0.969 ms
64 bytes from 192.168.146.129: icmp_seq=16 ttl=64 time=0.802 ms
64 bytes from 192.168.146.129: icmp_seq=17 ttl=64 time=1.11 ms
64 bytes from 192.168.146.129: icmp_seq=18 ttl=64 time=1.07 ms
64 bytes from 192.168.146.129: icmp_seq=19 ttl=64 time=1.15 ms
64 bytes from 192.168.146.129: icmp_seq=20 ttl=64 time=1.07 ms
```

Task 3: Vulnerable Web Application Deployment

- Screenshot of DVWA or Mutillidae homepage loaded in the browser



- URL bar visible showing the application address



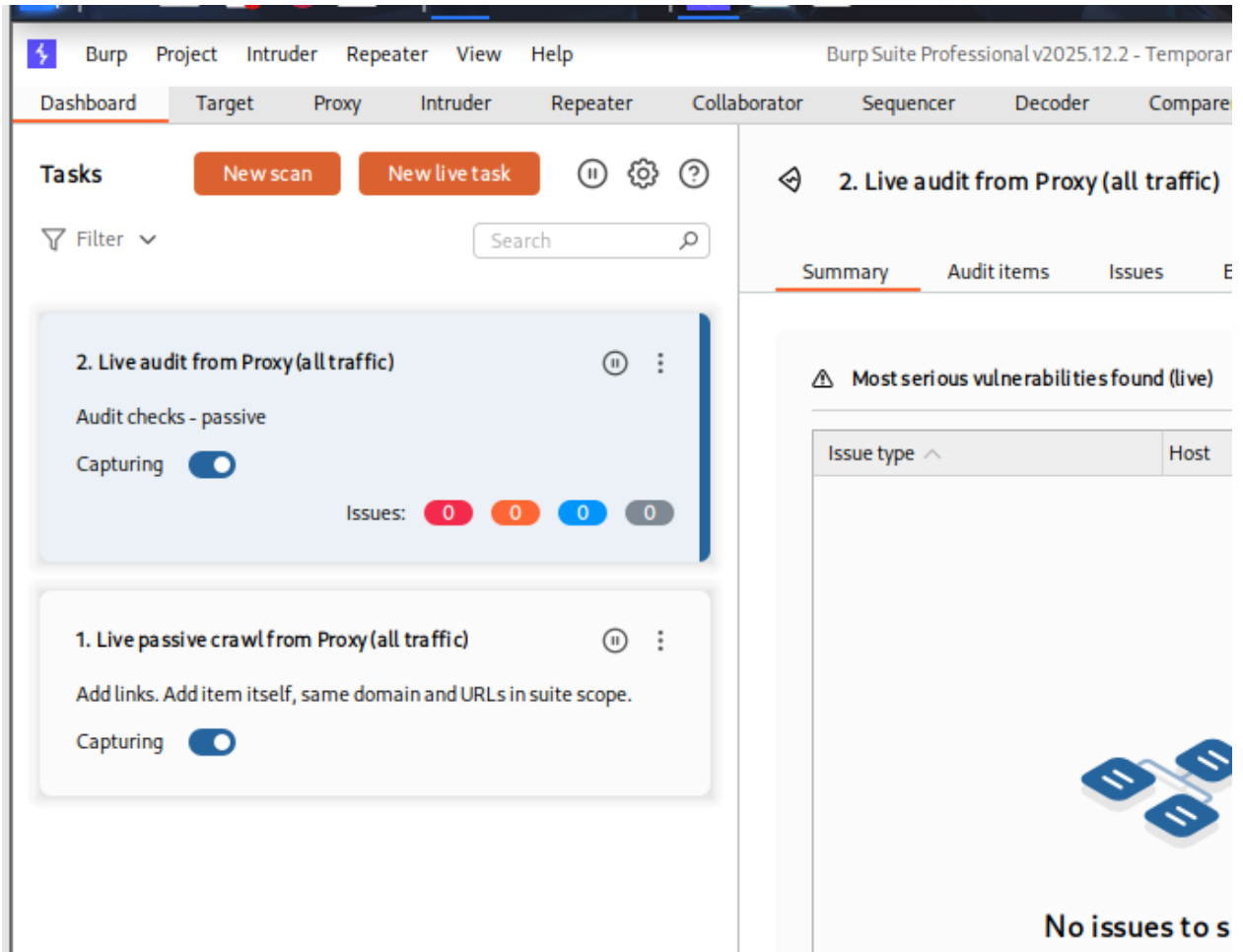
- Short

explanation of DVWA web application

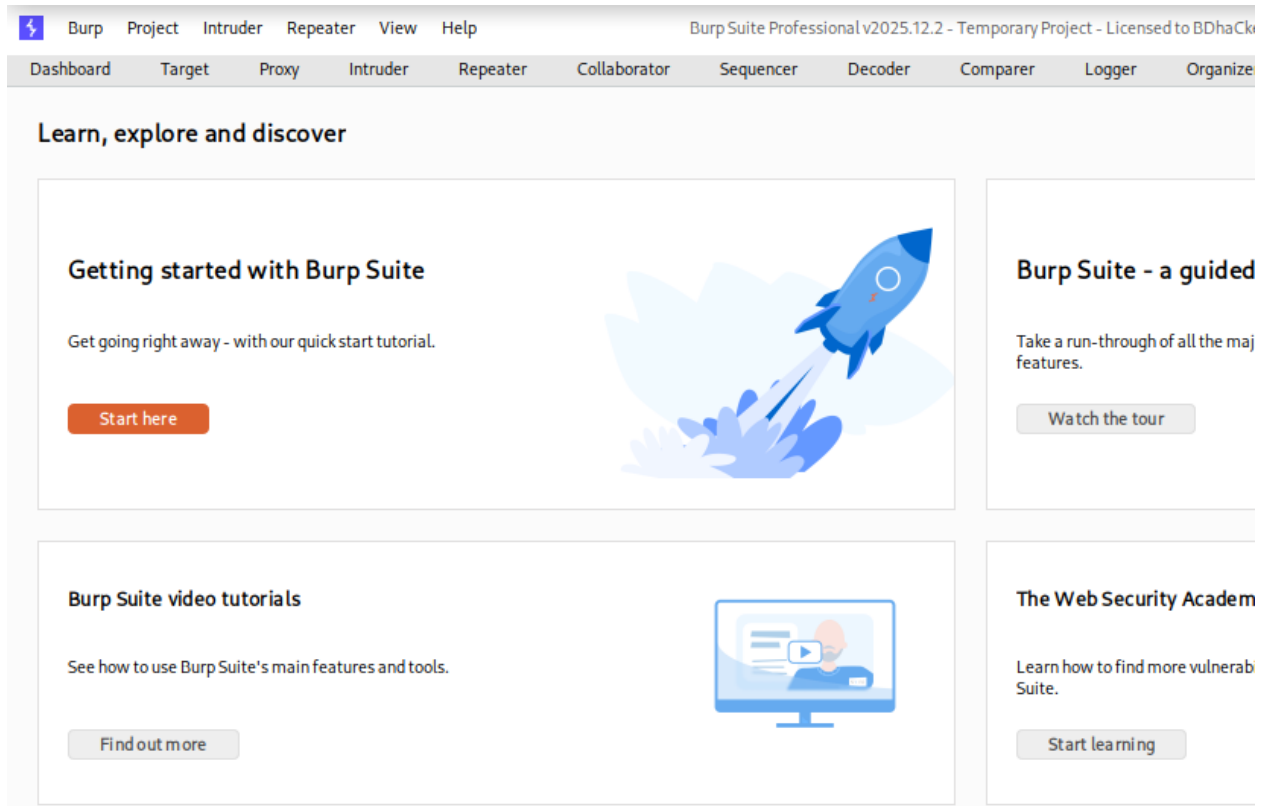
Damn Vulnerable Web Application (DVWA) was chosen for this lab because it is specifically designed to demonstrate common web application vulnerabilities in a controlled and educational environment. DVWA provides multiple security levels and includes well-known vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), and Command Injection. This makes it suitable for practicing web application testing techniques while clearly illustrating how insecure coding practices can be exploited and how proper security controls can mitigate these risks.

Task 4: Burp Suite Professional Installation & Licensing

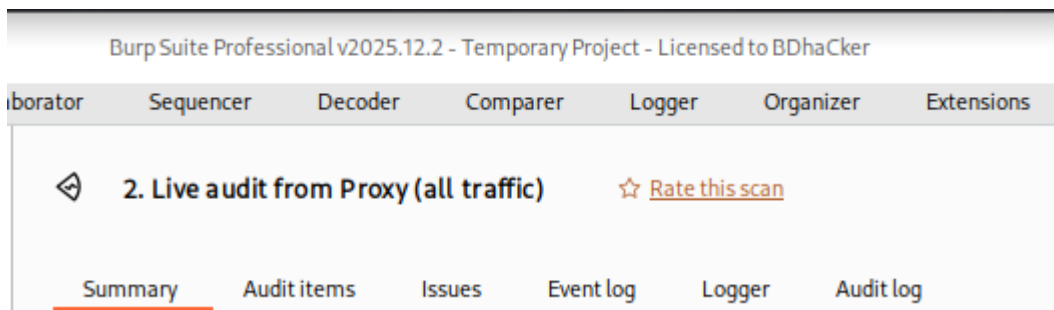
- Screenshot of Burp Suite Professional dashboard



- Screenshot showing Burp Suite running without trial limitations



- Screenshot of license status (no errors or expiration warnings)



- Brief explanation of the licensing process

the Burp Suite licensing process is the way I activate the Professional version so I can use advanced web application testing features during my labs and practical exercises. After purchasing a student license from PortSwigger, I receive a license key.

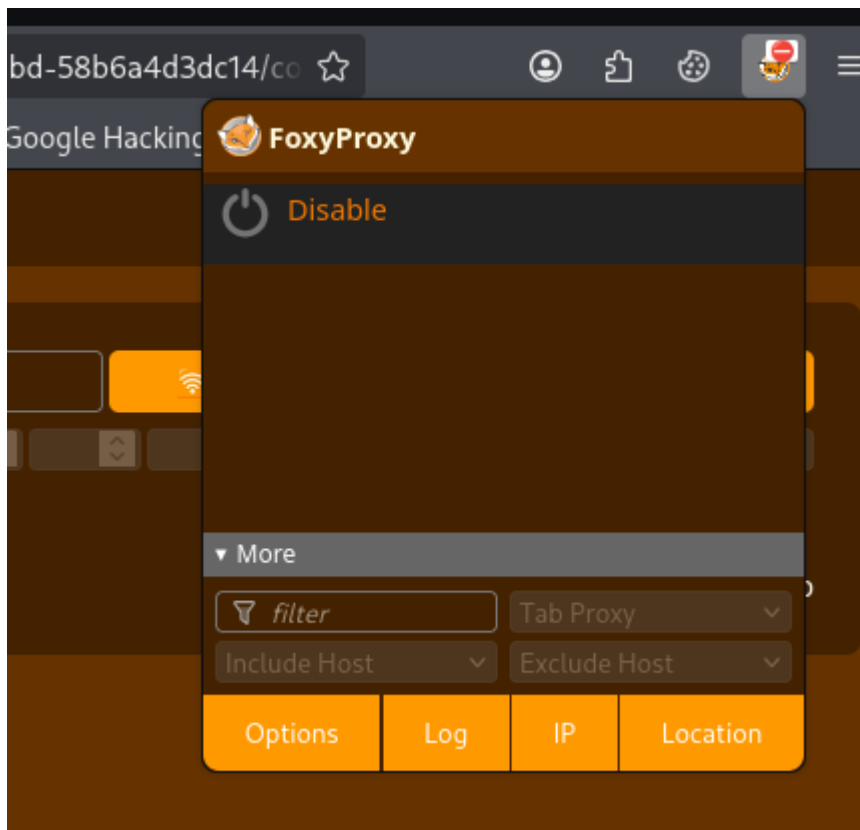
I enter this license key into Burp Suite, where the software connects to PortSwigger's licensing server to verify it. Once the license is validated, premium features such as the web vulnerability scanner, advanced Intruder options, and automated testing tools become available for my coursework and hands-on practice.

If internet access is limited in a lab environment, Burp Suite also supports offline/manual activation by generating a request file and uploading it to the PortSwigger licensing portal to get an activation response.

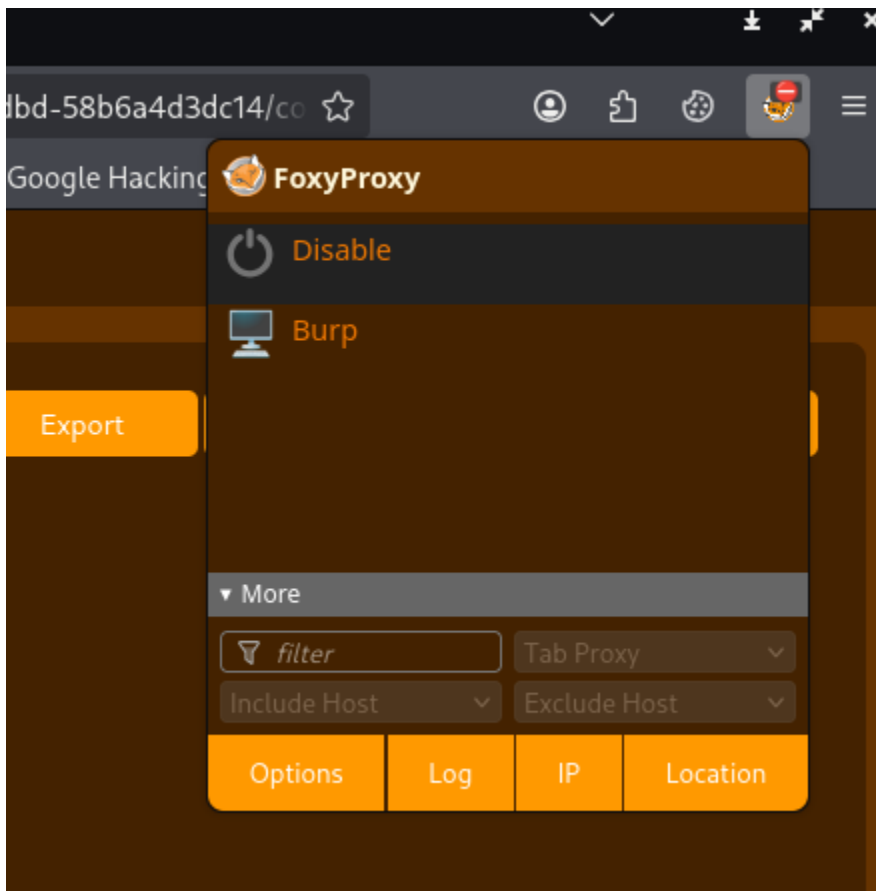
Overall, the licensing process allows me, as a student, to legally access Burp Suite's professional tools for learning and improving my web application security skills.

Task 5: Browser Proxy & Interception Configuration

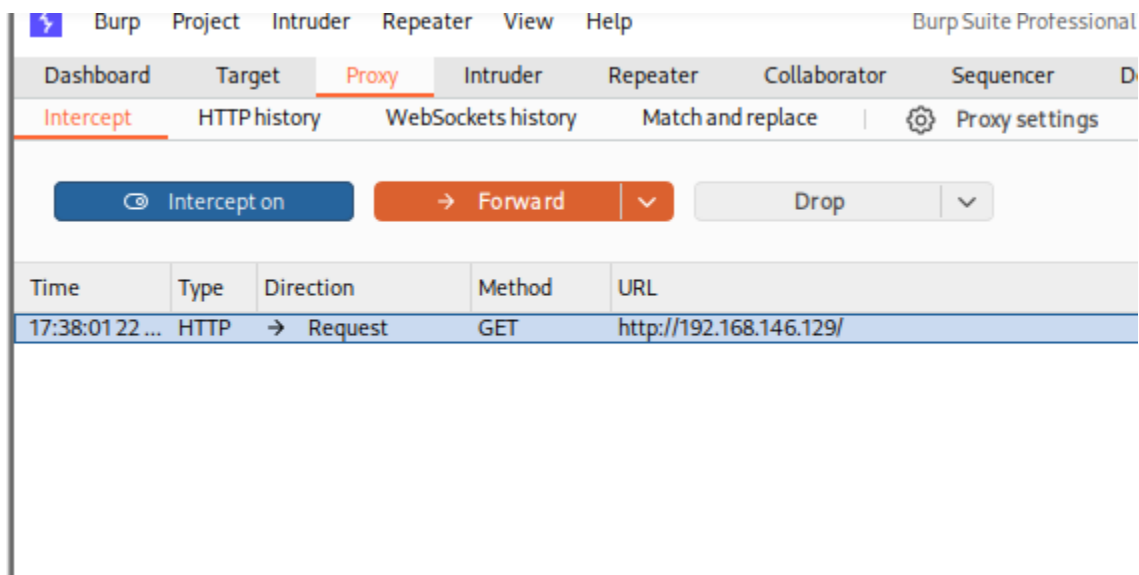
- Screenshot of FoxyProxy installed in Firefox



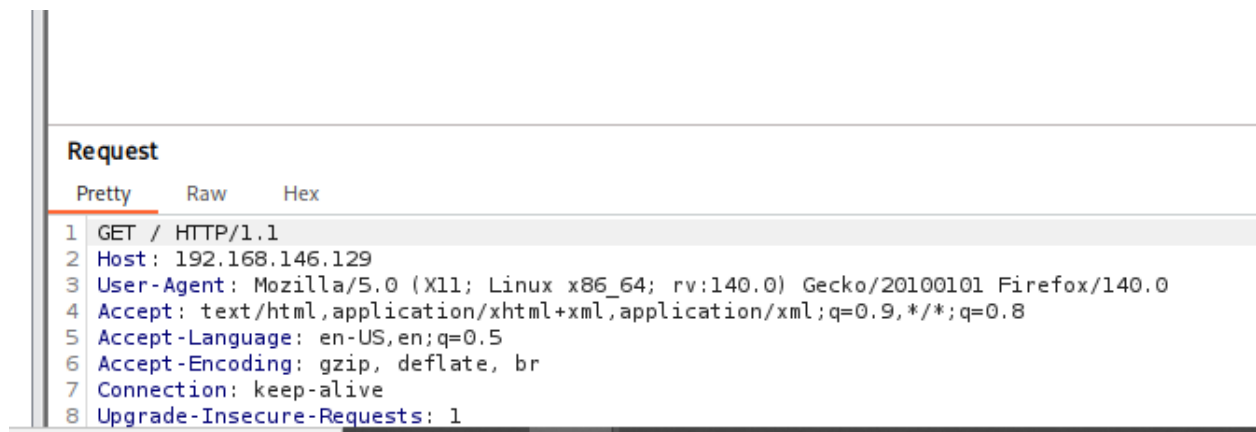
- Screenshot showing FoxyProxy switched ON and pointing to Burp



- Screenshot of Burp Suite Proxy → Intercept tab capturing HTTP requests



- Screenshot of browser request being intercepted and paused in Burp



- Short explanation of how interception works

Burp Suite Intercept works by acting as a man-in-the-middle proxy between the browser and the web application. When Intercept is turned on, all HTTP and HTTPS requests sent from the browser are paused before reaching the server. This allows me, as a student, to view and modify requests (such as parameters, headers, or cookies) before forwarding them. Once the request is sent, the server's response can also be intercepted and analyzed, helping me understand how web applications communicate and identify security vulnerabilities.

Conclusion

This lab successfully demonstrated the setup and configuration of a basic offensive security testing environment. By installing Metasploitable2 and Kali Linux within a virtualized platform, network connectivity between the attacker and target machines was verified using IP identification and ping tests. A vulnerable web application (DVWA or Mutillidae) was deployed and accessed to confirm proper service availability.

Additionally, Burp Suite Professional was correctly installed and licensed, and Firefox was configured with FoxyProxy to enable web traffic interception. The successful interception of HTTP requests confirmed proper proxy configuration and validated the ability to analyze client-server communication. Overall, this lab achieved its objectives by providing practical experience in environment setup, connectivity verification, vulnerable application deployment, and professional web proxy configuration, reinforcing essential skills required for web application security testing.

